

Supplementation of herbal plants differently modulated metabolic profile, insulin sensitivity, and oxidative stress in transition dairy cows fed various extruded oil seeds

F. Hashemzadeh-Cigari^{a,*}, G.R. Ghorbani^a, M. Khorvash^a, A. Riasi^a,
A. Taghizadeh^b, Q. Zebeli^c

^a Department of Animal Science, Isfahan University of Technology, Isfahan 84156, Iran

^b Department of Animal Science, University of Tabriz, Tabriz, Iran

^c Institute of Animal Nutrition and Functional Plant Compounds, Vetmeduni, 1210 Vienna, Austria

ARTICLE INFO

Article history:

Received 7 October 2013

Received in revised form 16 April 2014

Accepted 23 October 2014

Keywords:

Herbal mixture

Fatty acid sources

Glucose metabolism

Oxidative status

Transition period

ABSTRACT

The study investigated the effects of a mixture of herbal plants (HM) and two sources of unsaturated fatty acids (FA), extruded linseed (LS) and soybean (SB), on metabolic profile, insulin sensitivity, and oxidative status of transition dairy cows. Thirty-two prepartum Holstein cows, blocked by parity and calving day, were randomly assigned to 1 of 4 treatments, in a 2 × 2 factorial design, starting from 25 days before the expected calving date to 26 days postpartum. The supplementation rates of HM were 150 and 170 g/animal/day at pre- and postpartum, respectively. Blood samples were analyzed for metabolites on day 7.15 ± 1.70 prepartum and on days 1 and 21 postpartum. An intravenous glucose tolerance test (IV-GTT) was conducted on day 25 postpartum. Data showed that cows supplemented with HM had lower serum concentration of NEFA (0.395 vs. 0.602 ± 0.044 mmol/L; *P* < 0.01) and NEFA to insulin ratio (*P* < 0.01) postpartum. Compared to animals fed SB-based diets, cows fed the LS-based diet had greater serum glucose concentration during prepartum (80.7 vs. 71.3 ± 3.32 mg/dL; *P* = 0.06) and postpartum period (86.3 vs. 73.5 ± 3.35 mg/dL; *P* = 0.01), as well as lower NEFA (0.425 vs. 0.572 ± 0.044 mmol/L; *P* = 0.03) and insulin to glucose ratio (*P* < 0.01) postpartum. Revised quantitative insulin-sensitivity check index revealed that supplementing HM in LS-based diet improved insulin sensitivity (0.45 vs. 0.41 ± 0.013; *P* = 0.03) prepartum, whereas after parturition, the HM addition was effective for both oil seeds (0.40 vs. 0.37 ± 0.008; *P* = 0.06) in enhancing insulin sensitivity. Result of IV-GTT indicated that cows fed LS-based diets had higher basal glucose concentration (63.7 vs. 55.7 ± 2.37; mg/dL; *P* = 0.02) and lower glucose area under the curve (995.8 vs. 1529.5 ± 100.7; mg/dL × 45 min; *P* < 0.01). Supplementing HM resulted in greater total antioxidant capacity prepartum (0.55 vs. 0.48 ± 0.017 nmol/L; *P* = 0.01) and lower malondialdehyde concentration at prepartum (1.03 vs. 1.96 ± 0.140 μmol/L; *P* < 0.01) and postpartum (1.32 vs. 1.88 ± 0.178 μmol/L; *P* = 0.04). Although feeding LS ameliorated insulin resistance, this feeding strategy lowered total antioxidant capacity prepartum (0.48 vs. 0.55 ± 0.017 nmol/L; *P* < 0.01) and increased malondialdehyde concentration postpartum more than the SB diet (1.91 vs. 1.28 ± 0.172 μmol/L; *P* = 0.02). Overall, both HM supplementation and LS feeding improved metabolic profile and insulin response following glucose infusion, although feeding of LS-based diets induced an increased oxidative stress.

© 2014 Elsevier B.V. All rights reserved.

* Corresponding author.

E-mail addresses: f.hash_1362@yahoo.com, hashemzadeh.farzad@gmail.com (F. Hashemzadeh-Cigari).

1. Introduction

The transition period is the most critical stage for health and subsequent performance of dairy cows, which are exposed to severe physiological adaptations (Drackley, 1999). Insulin resistance develops during the periparturient period and increases the availability of glucose, amino acids and fatty acids (FA) to mammary gland metabolism (Bell, 1995), but exaggerated insulin resistance during late pregnancy and early lactation could promote greater mobilization of body fat reserves and expose transition cows to metabolic dysfunctions (Hayirli et al., 2011). Therefore, dietary management strategies should be considered to ameliorate insulin resistance and optimize glucose and lipid metabolism.

Beside the role of FA as a source of energy to alleviate the negative energy balance in early lactation, these biologically active molecules may regulate biological processes in various tissues (Mashek and Grummer, 2003). It has been reported that abomasal infusion of linseed oil showed insulin-sensitizing as well as antilipolytic effects in non-lactating cows (Pires et al., 2008). Also, feeding whole flaxseed to transition dairy cows resulted in a better performance, improved blood metabolic profile and liver function (Lessard et al., 2004; Zachut et al., 2011). However, enriching diets with omega-3-rich FA (n-3 FAs), such as by supplementing linseed, may concurrently predispose cows to experience more symptoms of oxidative stress and subsequently disturb metabolic homeostasis (Gobert et al., 2009). Some researchers reported that n-3 FA supplementation to fattening steers (Scislowski et al., 2005) or rabbits (Tres et al., 2009) led to greater oxidative stress than omega-6 FA (n-6 FA) supplementation, but the effects of different FA on oxidative status of transition dairy cows that normally experience physiologic oxidative stress remains unclear (Turk et al., 2008).

Herbal plants are used for many purposes such as improvement of physiological functions or for prevention and treatment of certain pathological conditions (Frankic et al., 2009). Many *in vitro* and *in vivo* non-ruminant models have demonstrated that herbs such as cinnamon bark, clove bud and turmeric root, which used in the present study, improve insulin resistance and lipid metabolism (Mahfouz, 2011; Akilen et al., 2012; Kuroda et al., 2012) and act as a potent immunomodulatory, hepatoprotective as well as antioxidant agents (Miguel, 2010). Results by Tedesco et al. (2004) also suggest that transition cows supplemented with silymarin, a natural hepatoprotector, showed an enhanced lactation performance and body condition score. In a series of studies, plant polyphenol extracts, supplied in association with vitamin E in diets enriched in n-3 FA, were able to reduce lipoperoxidation and ameliorate oxidative stress of sheep and dairy cows (Gladine et al., 2007; Gobert et al., 2009).

Previous studies have demonstrated that synergistic interactions of bioactive compounds in herbal plants are mainly responsible for their potent health beneficial activities such as antioxidant and anticancer properties (Liu, 2004). It is well known that there is no single antioxidant with capability of scavenging all kinds of

free radicals or inhibitory activities of lipid peroxidation (Moure et al., 2001). Therefore, combinations of antioxidants with possible synergism are preferred for preventing free-radical-induced disorders (Moure et al., 2001). Our preliminary experiment showed that quaternary mixtures of herbs, which are used in the present study, exhibited synergistic antioxidant activity (+29%) relative to the activity of each single herb (unpublished data). It suggests that the combining of herbs potentiates positive response of each single extract, leading to an increased antioxidant activity due to a potential additive effect.

Although the body of evidence evaluating the effects of various kinds of FAs on glucose and insulin metabolism in non-lactating dairy cows is increasing, information on the effects of enriching diets with n-3 or n-6 FA in the transition period is still limited. Also, there are limited data in the literature on the effects of supplementing herbal plants as a multi-functional additive in transition period. With regard to some positive impact of feeding n-3 FA enriched diets and herbal plants as mentioned earlier, it was hypothesized that administering these two compounds to transition dairy cows may have additive or synergistic effects on optimizing metabolic health.

The objective of this study was to investigate whether supplementation of a specific blend of several herbal plants could improve glucose metabolism, antioxidant status and metabolic status of transition dairy cows, which were offered two different extruded oil seeds including linseed or soybean.

2. Materials and methods

This article is a part of a larger study evaluating the effects of supplementing an HM with antioxidant activity on performance, potential amelioration of glucose metabolism, insulin sensitivity and oxidative stress status in periparturient dairy cows fed two different oil seed sources. The present article focuses on the treatment effects on metabolic profile, glucose and insulin responsiveness to glucose challenge, lipid and protein metabolism and non-enzymatic antioxidant status of transition cows, whereas data of intake, lactational performance in more detail, nutrient digestibility, and ruminal fermentation are presented in the companion paper (Hashemzadeh et al., unpublished data).

2.1. Animals, diets, and treatments

The experiment was conducted at the Dairy Facilities of the Lavark Research Station (Isfahan University of Technology, Isfahan, Iran) from December 2011 to May 2012 using 20 multiparous (mean $\pm$ SD, BW = 727 $\pm$ 22.3 kg; BCS = 3.60 $\pm$ 0.135) and 12 primiparous (mean $\pm$ SD, BW = 608 $\pm$ 15.9 kg; BCS = 3.23 $\pm$ 0.107) dry Holstein cows.

All animals were managed and experimental manipulations performed in accordance with protocol (no. 19293) approved by the Iranian Council of Animal Care (1995). Cows were stratified by parity (primiparous or multiparous) and similar predicted calving date and randomly assigned to 1 of 4 treatment groups in a 2 $\times$ 2 factorial

design of n-3 enriched diet (LS; using extruded linseed as a n-3 FA source) or n-6 enriched diet (SB; using extruded soybean as main n-6 FA source) combined with or without herbal mixture (HM). Cows were housed in individual stalls from 30 days prepartum and received treatments from 25 days before the expected calving date and through 26 days postpartum. Cows were fed total mixed rations (TMR), formulated using CNCPS model software, once daily prepartum and twice daily postpartum ad libitum (allowing 5–10% orts) with free access to water. Ingredients (DM basis) of prepartum and postpartum diets and their chemical and FA composition are shown in Table 1. Extruded linseed (Nutex, Interagri-Dumoulin, Seilles, Belgium) consisted in an extruded mixture of linseed, wheat, sunflower cake, field beans, peas, butylated hydroxytoluene (BHT) and salt (584, 150, 140, 51, 50, 20 and 5 g/kg total raw materials, respectively). Total fat content of extruded linseed was 257 g/kg of DM. n-3 and n-6 FA contents were 513 and 170 g/kg of total FAs, respectively. Extruded soybeans contained 187 g/kg of DM total fat which contained of 482 g/kg n-6 fatty acids. Dried cinnamon barks (*Cinnamomum zeylanicum*), Turmeric roots (*Curcuma longa*), rosemary leaves (*Rosemarinus officinalis*) and clove buds (*Eugenia caryophyllata* Thunb.) were purchased in a single lot from local market in Isfahan, Iran. Herbal mixture were ground through a 1-mm screen using a Wiley mill and mixed in the proportion of 60% rosemary, 18% cinnamon bark, 18% turmeric and 4% clove bud, weekly. Total phenolic contents were 58.8, 31.3, 77.7 and 110.2 g tannic acid equivalent per kg for cinnamon bark, turmeric root, rosemary leaves and clove bud, respectively, that provided 67.169 g tannic acid equivalent phenolic compounds per kg of herbal mixture (HM). Supplementation rate of HM in the diet was 150 and 170 g/animal/day at pre- and postpartum, respectively. The powder of herbal material was top-dressed and mixed with a portion of the TMR, and attention was paid that most of the HM was ingested.

2.2. Sample collection and analyses

Amounts of feed offered and refused were recorded daily for individual cows. The offered feeds and refusals were sampled weekly and composited every other week. Dry matter of feed samples were determined after oven-drying for 72 h at 55 °C and then ground through a 1-mm screen using a Wiley mill (Arthur Thomas Co., Philadelphia, PA, USA). The ground samples were analyzed for N using the Kjeldahl (AOAC, 1990), ether extract (AOAC, 1990), ash (AOAC, 1990). The neutral detergent fiber (NDF) and acid detergent fiber (ADF) were determined using the methods described by Van Soest et al. (1991), with heat stable α -amylase used in the NDF procedure. Dietary fatty acids extracted and methylated as described by O'Fallon et al. (2007) and quantified by gas chromatography as described by Goli et al. (2013).

Following parturition, cows were milked 3 times daily at 0200, 1000, and 1800 h in the milking parlor and milk yields were recorded every other day. Milk samples were collected on day 8, 16, and 24 postpartum from 3 consecutive milkings and preserved using potassium dichromate. Milk samples were analyzed for fat, protein, lactose, and total

Table 1

Ingredients and nutrient composition (DM basis) of prepartum and postpartum diets, including linseed-based diet (n-3-rich) and a soybean-based (n-6-rich) diet.

Oil seed sources	Prepartum ^a		Postpartum ^b	
	Linseed	Soybean	Linseed	Soybean
Ingredients (% of DM)				
Alfalfa hay	17.6	18.3	20.2	20.8
Corn silage	30.5	31.4	19.6	20.2
Wheat straw	5.4	5.9	–	–
Barley grain	11.7	12.2	16.4	16.9
Corn grain	11.7	12.2	16.1	16.6
Fishmeal	0.4	0.4	2.0	0.7
Extruded linseed ^c	8.5	–	9.6	–
Extruded soybean	–	12.3	–	14.0
Soybean meal	9.7	3.1	12.0	6.6
Calcium–carbonate	1.0	1.0	0.8	0.8
Dicalcium–phosphate	0.6	0.6	0.3	0.3
Anionic/vitamin/mineral supplement ^d	2.5	2.5	–	–
Mineral supplement ^e	–	–	0.8	0.8
Vitamin supplement ^f	–	–	0.5	0.5
Salt	–	–	0.4	0.4
Sodium bicarbonate	–	–	1.2	1.2
Chemical composition (% of DM, unless otherwise stated)				
DM (% as fed basis)	53.5	52.9	62.0	61.7
OM	90.8	91.7	90.0	91.9
CP	13.7	14.2	16.9	17.3
NDF	41.8	42.1	36.1	35.8
ADF	21.5	20.9	17.3	17.3
EE	4.4	4.5	5.1	5.3
NE _t ^g	1.45	1.46	1.61	1.62
NFC ^h	31.4	32.1	33.8	34.3
Fatty acids (% of total FA)				
C14:0	0.42	0.31	0.40	0.28
C16:0	18.7	18.1	19.3	18.4
C18:0	5.6	5.1	5.7	5.4
C18:1	25.7	22.2	26.8	28.0
C18:2	25.8	41.5	27.5	38.1
C18:3	19.2	8.4	16.9	7.4
n-6/n-3	1.34	4.97	1.62	5.21

^a From 25 days before calving until calving.

^b From calving until 26 days of lactation.

^c Extruded linseed (Nutex, Interagri-Dumoulin, Seilles, Belgium), comprised of an extruded mixture of linseed, wheat, sunflower cake, field beans, peas, BHT and salt (584, 150, 140, 51, 50, 20 and 5 g/kg total raw materials, respectively).

^d Contained: vitamin A, 250,000 IU/kg; Vitamin D, 40,000 IU/kg; Vitamin E, 2000 IU/kg; biotin 20 mg/kg; Ca, 150 g/kg; Cl, 136 g/kg; Mg, 40 g/kg; S, 30 g/kg; Mn 800 mg/kg; Zn, 800 mg/kg; Cu, 400 mg/kg; Co, 4 mg/kg; I, 12 mg/kg; Se, 7 mg/kg; Cr, 14 mg/kg.

^e Contained: Co, 120 mg/kg; Cu, 4000 mg/kg; I, 150 mg/kg; Fe, 800 mg/kg; Mn, 10,000 mg/kg; Se, 80 mg/kg; Zn, 16,000 mg/kg.

^f Contained: Vitamin A, 1,300,000 IU/kg; Vitamin D, 360,000 IU/kg; Vitamin E, 12,000 IU/kg.

^g Calculated according to Cornell Net Carbohydrate and Protein System (CNCPS).

^h NFC calculated as 100 – (% NDF + % CP + % ether extract + % ash) (NRC, 2001).

solid by an automated near-infrared spectroscopy analyzer (Milko-Scan, 134 BN, Foss Electric, Hillerød, Denmark). Body condition score was recorded at the beginning of the experiment, calving day and day 24 postpartum using a scale from 1 to 5, including 0.25 points, where 1 = thin and 5 = extremely fat. Energy balance estimates for pre- and postpartum periods were calculated as described by Hayirli et al. (2011). Herbal materials were powdered in a coffee grinder. Plant extracts preparation and measurement of total phenolic compounds, with the Folin

& Ciocalteu's phenol reagent, were conducted using the methods described by Bamdad et al. (2006).

Blood samples were collected approximately 3.5 h after the morning feeding from the coccygeal vein using evacuated serum separator tubes containing clot activator on day 7.15 ± 1.70 before expected calving date and on day 1 and day 21 postpartum. All tubes were centrifuged at $1850 \times g$ for 20 min. The serum samples were frozen (-20°C) until analyses. The concentration of serum glucose, cholesterol, blood urea-N, high density lipoprotein (HDL), urea, triglyceride (TG), total protein, albumin, creatinine, alanine aminotransferase (ALT), aspartate aminotransferase (AST), γ -glutamyltransferase (GGT), alkaline phosphatase (ALP) were measured by an autoanalyzer (Abbott Alycon 300, USA) using commercial kits (Pars Azmoon Co., Tehran, Iran) according to the manufacturer's instructions. Before application, the autoanalyzer was calibrated with the control sera N and P (TrueLab N[®] and TrueLab P[®], respectively; Pars Azmoon Co., Iran) and a calibrator solution (TrueCal U[®], Pars Azmoon Co., Iran). Total antioxidant capacity, β -hydroxy-butyrate (BHBA) and non-esterified fatty acid (NEFA) of sera were determined by commercial colorimetric kits (Randox Laboratories Ltd., Ardmore, UK) using the same autoanalyzer. Globulin concentrations were obtained by deducting albumin from total protein. Low density lipoprotein (LDL) concentration estimated using Friedewald equation (Friedewald et al., 1972). Malondialdehyde (MDA) levels were measured using the thiobarbituric acid reactive substances (TBARS) method previously described by (Meagher and FitzGerald, 2000). Serum insulin level was measured using enzyme linked immunosorbent assay kit (Monobind Inc., CA, USA). Intra and inter-assay coefficients of variation for measuring insulin were 6.9 and 8.2%, respectively. Copper-induced oxidation of serum was studied and conjugated dienes (CD) concentration was measured by method described by Schnitzer et al. (1998). Basal and maximum absorbance at 243 nm and the net area under the curve were calculated and treated as the final result.

The revised quantitative insulin-sensitivity check index (RQUICKI), a relative insulin sensitivity measure, was estimated as $\text{RQUICKI} = 1/[\log(\text{glucose}) + \log(\text{insulin}) + \log(\text{NEFA})]$ according to Holtenius and Holtenius (2007).

2.3. Catheterization and i.v. glucose tolerance test

On day 25 postpartum, skin area over the jugular vein was clipped and anesthetized with 4 mL of lidocaine HCl (2%, no epinephrine; Pasteur Institute, Iran) and an indwelling sterile and nonpyrogenic catheter (6 French scale, 100 cm long; Supa Co, Karaj, Iran) was inserted through a 12-gauge sterile needle until approximately 20 cm of tubing was inside the jugular vein. The needle was subsequently removed. The catheter was flushed with a 5% sodium citrate solution to maintain patency. The next morning, feed was removed at 0700 h. Water was available during i.v. glucose tolerance test (GTT). At 0900 h, the IV-GTT was conducted. Blood was collected at $-5, 0, 2, 5, 10, 15, 30, 45, 60$ and 90 min relative to i.v. infusion of 150 mg dextrose (50% wt/vol)/kg BW, followed by 50 mL of sterile saline to flush catheters and to prevent contamination

of subsequent samples. Dextrose solution was warmed to body temperature before injection. Blood samples were collected to heparinized tubes and handled as previously described.

Pharmacokinetic parameters in response to IV-GTT for glucose and insulin including basal levels, relative clearance rate (CR; 2–45 min), time to half-maximal concentration ($T_{1/2}$), time to reach basal level (T_{basal}) calculated as described previously (Opsomer et al., 1999). Incremental area under the curve (AUC; 0–45 min) was calculated using the trapezoidal rule with baseline subtraction. Basal glucose and insulin concentrations were calculated by averaging values from samples taken 5 and 0 min before IV-GTT. The CR (%/min) is defined as glucose disappearance rate and illustrates the clearance of the infused glucose. Time to half-maximal concentration (min) is defined as the time required for the glucose concentration to fall by one-half, and it hence illustrates the clearance of the infused glucose.

2.4. Statistical analyses

The blood metabolites data were analyzed for day -7 as prepartum data and days $+1$ and $+21$ were considered as postpartum data using the MIXED procedure of SAS (SAS 9.0 for windows, SAS Institute Inc., Cary, NC, USA). The model for prepartum included the fixed effects of feeding treatments (different oil seeds, HM and their interactions), parity (primiparous, multiparous) and the random effect of cow nested within treatment and the residual error. The model for postpartum included, fixed effects of feeding treatments (different oil seeds, HM and their interactions), parity (primiparous, multiparous), sampling day and the random effect of cow nested within treatment and the residual error. Also, interactions between treatments and sampling times were included in the model, if they were significant. Data were presented as least square means (LSM) $\pm$ standard error of the mean (SEM). A threshold of significance was set at $P < 0.05$; trends were declared at $0.05 < P < 0.10$.

3. Results

Dry matter intake was greater for prepartum cows fed LS diet than SB enriched diets ($P < 0.01$; Table 2). After parturition, cows fed LS diet had greater DMI ($P < 0.01$) and lost less BCS than SB supplemented cows ($P < 0.01$), and the energy balance was less negative in this group ($P = 0.04$). During the prepartum period, HM-supplemented cows tended ($P = 0.10$) to have greater DMI than unsupplemented cows. There was an interaction ($P < 0.01$) between oil seeds sources and HM supplementation on DMI of postpartum cows, whereby DMI increased when HM was added to the SB diet ($P < 0.01$; Table 2) but remained unchanged in LS diets ($P > 0.85$). Milk production was greater for cows fed LS diet than cows fed SB ($P = 0.05$). Milk lactose concentration increased with LS ($P = 0.02$); however, no significant effect on other milk constituents was observed. A tendency was observed for increased milk yield ($P = 0.08$) and milk fat concentration ($P = 0.06$) in cows fed HM vs. control cows. As expected, multiparous cows consumed greater DM at prepartum (15.5 vs. 12.8 kg/d; $P < 0.01$) and postpartum

Table 2

Effects of supplementing with (HM) or without herbal mixture (NO) and different oil seed sources (OS; linseed or soybean) to transition dairy cows on pre-partum and post-partum DM and energy intake, BW, BCS, calf weight and energy balance.

Item	Linseed		Soybean		SEM ^a	P values		
	HM	NO	HM	NO		OS	HM	OS × HM
Pre-partum								
DMI (kg/d)	15.5	15.3	13.7	12.5	0.4	<0.01	0.10	0.25
Energy balance ^b	7.7	7.6	5.4	4.2	0.7	<0.01	0.34	0.45
Post-partum								
DMI (kg/d)	20.4	20.3	18.7	15.6	0.4	<0.01	0.01	<0.01
BCS change (%) ^b	−19.2	−19.0	−24.7	−30.3	2.1	<0.01	0.19	0.16
Energy balance ^c	−8.2	−8.5	−12.7	−12.8	1.2	0.04	0.93	0.96
Milk (kg/d)	38.5	38.2	37.8	32.9	1.5	0.05	0.08	0.13
Milk component								
Fat (%)	4.37	4.26	4.76	4.23	0.17	0.30	0.06	0.23
Protein (%)	3.15	3.13	3.09	3.05	0.08	0.40	0.66	0.83
Lactose (%)	5.73	5.77	5.53	5.44	0.11	0.02	0.79	0.49
Total solids (%)	13.23	13.33	14.42	12.92	0.22	0.60	0.36	0.17

^a Standard error of the means.

^b Percent of BCS change from day 1–24 postpartum relative to parturition BCS.

^c Prepartum and post-partum energy balance calculated as described by Hayirli et al. (2011).

(21.2 vs. 16.3 kg/d; $P < 0.01$) period and had more milk yield (42.0 vs. 31.4 kg/d; $P < 0.01$) than primiparous cows. Also, multiparous cows tended to experience a more severe negative energy balance (−12.5 vs. −8.6 Mcal/d; $P = 0.06$) than primiparous cows at postpartum period.

Results of serum metabolic profile are presented in Table 3. No HM by oil seed sources interactions were observed for glucose, NEFA, BHBA and insulin concentration at prepartum ($P = 0.75$, 0.19, 0.32 and 0.71, respectively) and postpartum ($P = 0.21$, 0.24, 0.75 and 0.96, respectively; Table 3) period. There was an interaction between HM and oil seed sources ($P = 0.05$; Table 3) on NEFA to insulin ratio at prepartum period, as supplementing HM resulted in lower NEFA to insulin ratio in cows fed LS-based diet ($P = 0.04$) but not in SB-fed cows ($P = 0.41$). Supplementing HM did not affect insulin, insulin to glucose ratio and BHBA concentration at prepartum ($P = 0.53$, 0.83 and 0.52, respectively; Table 3) and postpartum ($P = 0.46$, 0.23 and 0.50, respectively) period. Cows fed HM showed lower serum concentration of NEFA ($P < 0.01$) and NEFA to insulin ratio ($P < 0.01$) during the postpartum period. In the current study, cows receiving LS-diet had greater plasma glucose ($P = 0.02$) and a lower NEFA ($P = 0.03$), insulin to glucose ratios ($P < 0.01$), and NEFA to insulin ratios ($P = 0.10$; Table 3) compared with cows receiving the SB diet during the postpartum period (Table 3). Prepartum glucose (79.5 vs. 71.5; $P = 0.07$) and postpartum BHBA (1.08 vs. 0.89; $P = 0.05$) and NEFA (0.51 vs. 0.48; $P = 0.07$) concentrations were greater in multiparous than primiparous cows.

The RQUICKI calculation was used in this experiment to describe differences in apparent insulin resistance of transition cows. There was an interaction ($P < 0.05$) between oil seed sources and HM on RQUICKI during the prepartal period. This interaction indicated an improvement of RQUICKI in cows supplemented with HM when cows fed LS-diet ($P = 0.03$), whereas this index was the same between cows fed HM in SB-diet ($P = 0.34$). During the postpartal period, cows supplemented with HM tended to have higher RQUICKI value ($P = 0.06$) than the unsupplemented cows (Table 3). There was a significant time effect ($P < 0.01$) on RQUICKI value (Fig. 1). Results indicate that 1 day

postpartum, cows experienced impaired insulin responsiveness compared to 7 days prepartum and 21 days postpartum. Also, primiparous cows showed greater prepartum RQUICKI value (0.44 vs. 0.41; $P = 0.04$) compared to multiparous cows.

There was an interaction ($P = 0.01$) between HM and oil seed sources on TG concentration during the prepartum since adding HM to LS-diet lowered TG ($P = 0.05$), but this response was not observed in cow fed the SB-diet ($P = 0.58$; Table 4). Postpartum NEFA to cholesterol ratio was lower ($P = 0.04$) in cows supplement with HM than unsupplemented cows. Lipid metabolism related variables were not affected by different oil seed sources.

There were interactions between HM and oil seed sources on prepartum albumin concentration ($P < 0.01$) and NEFA to albumin ratios ($P = 0.04$) because cows supplemented with HM in SB-diet had a higher albumin concentration ($P < 0.10$), but adding HM to cows fed LS-diet did not affect albumin concentration ($P = 0.98$). At prepartum period, cows supplemented with HM in LS-based diet had the lowest NEFA to albumin ratio than other treatments

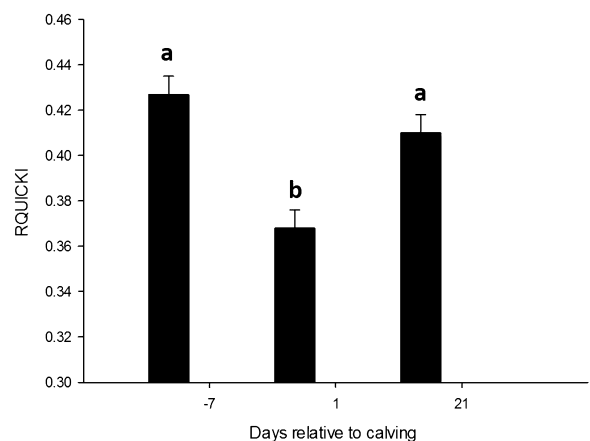


Fig. 1. Effects of time relative to calving on LSM ± SEM of revised quantitative insulin-sensitivity check index (RQUICKI).

Table 3

Effects of supplementing with (HM) or without herbal mixture (NO) and different oil seed sources (OS; linseed or soybean) to transition dairy cows on serum energy related metabolites.

Item	Day ^a	Linseed		Soybean		SEM ^a	P values		
		HM	NO	HM	NO		OS	HM	OS × HM
Glucose (mg/dL)	Pre	83.5	77.9	72.6	70.1	4.5	0.06	0.39	0.75
	Post	91.9	80.6	73.1	73.8	4.8	<0.01	0.28	0.21
NEFA (mmol/L) ^a	Pre	0.14	0.27	0.25	0.25	0.04	0.34	0.15	0.19
	Post	0.36	0.49	0.43	0.71	0.06	0.03	<0.01	0.24
BHBA (mmol/L) ^a	Pre	0.85	0.89	0.98	0.80	0.10	0.90	0.52	0.32
	Post	1.02	0.98	1.02	0.93	0.09	0.82	0.50	0.75
Insulin (μU/mL)	Pre	14.1	16.1	15.5	15.9	1.8	0.73	0.53	0.71
	Post	10.9	11.8	12.4	13.3	1.2	0.24	0.46	0.96
Insulin: glucose	Pre	3.39	3.31	3.77	4.09	0.59	0.32	0.83	0.71
	Post	2.16	2.61	3.06	3.17	0.22	0.01	0.23	0.46
NEFA: insulin	Pre	0.009	0.020	0.017	0.013	0.003	0.76	0.35	0.05
	Post	0.034	0.046	0.036	0.058	0.004	0.10	<0.01	0.22
RQUICKI ^a	Pre	0.45	0.41	0.41	0.43	0.013	0.65	0.36	0.03
	Post	0.41	0.39	0.39	0.36	0.014	0.14	0.06	0.87

^a Pre, prepartum (measurements made on day 7 prior to calving); Post, postpartum (overall means of day 1 and 21 postpartum); SEM, standard error of the means; NEFA, non-esterified fatty acids; BHBA, β-hydroxybutyrate; RQUICKI, Revised quantitative insulin-sensitivity check index.

(Table 4). Multiparous cows showed lower ratio of albumin to globulin compared to primiparous cows at pre- (0.67 vs. 0.94; $P < 0.01$) and postpartum (0.66 vs. 0.82; $P < 0.01$) period.

During the postpartum period, NEFA to albumin ratio decreased with HM supplementation ($P = 0.01$, Table 4) and tended to decrease in cows fed LS-diets ($P = 0.09$) compared with SB-diets. Prepartum concentrations of urea-N were lower in cows supplemented with HM ($P < 0.01$) and in cows fed LS-diet compared to SB-diet ($P < 0.01$).

Postpartum creatinine ($P = 0.07$) concentration was lower in cows fed n-3 based diet.

Table 5 shows the serum activity of circulating enzymes originating from the liver. Prepartum AST activity was greater in cows supplemented with HM ($P < 0.01$). Supplementation with HM lowered serum GGT activity compared to unsupplemented cows at postpartum period ($P = 0.03$).

Results related to non-enzymatic antioxidant status and susceptibility of serum to oxidation are shown in Table 6 and Fig. 2A and B. Interactions between HM and

Table 4

Effects of supplementing with (HM) or without herbal mixture (NO) and different oil seed sources (OS; linseed or soybean) to transition dairy cows on serum metabolites.

Item	Day ^a	Linseed		Soybean		SEM ^a	P values		
		HM	NO	HM	NO		OS	HM	OS × HM
Cholesterol (mg/dL)	Pre	129.5	151.5	146.9	147.1	13.8	0.78	0.44	0.45
	Post	175.1	172.6	170.5	162.2	15.6	0.63	0.73	0.85
Triglyceride (mg/dL)	Pre	26.2	33.4	27.2	24.3	1.6	0.03	0.30	0.01
	Post	17.7	21.1	20.4	22.4	2.2	0.36	0.23	0.73
HDL (mg/dL) ^a	Pre	64.6	73.1	71.3	74.2	5.1	0.65	0.29	0.60
	Post	94.3	91.7	84.9	83.8	8.1	0.30	0.81	0.92
LDL (mg/dL) ^a	Pre	59.6	71.7	70.1	67.9	10.1	0.90	0.63	0.49
	Post	77.2	76.7	81.4	73.9	9.7	0.94	0.68	0.72
Total protein (g/dL)	Pre	8.7	8.4	9.1	8.5	0.29	0.45	0.22	0.68
	Post	9.5	8.8	9.5	9.4	0.26	0.22	0.14	0.26
Albumin (g/dL)	Pre	3.8	3.9	3.9	3.4	0.07	0.01	0.01	0.01
	Post	4.0	3.8	4.1	3.8	0.11	0.88	0.13	0.84
Globulin (g/dL)	Pre	4.8	4.5	5.1	5.1	0.27	0.13	0.59	0.6
	Post	5.5	4.9	5.5	5.6	0.23	0.15	0.34	0.17
Albumin: globulin (g/g)	Pre	0.82	0.92	0.79	0.65	0.04	0.01	0.91	0.05
	Post	0.74	0.80	0.74	0.71	0.03	0.16	0.65	0.24
Urea-N (mg/dL)	Pre	13.5	16.8	20.2	22.8	0.96	0.01	0.01	0.68
	Post	21.2	18.3	19.7	20.0	1.59	0.92	0.42	0.32
Creatinine (mg/dL)	Pre	1.45	1.46	1.60	1.53	0.29	0.75	0.72	0.72
	Post	1.30	1.32	1.40	1.69	0.12	0.07	0.21	0.27
NEFA: cholesterol	Pre	0.043	0.077	0.066	0.066	0.014	0.70	0.29	0.29
	Post	0.097	0.151	0.119	0.180	0.026	0.34	0.04	0.89
NEFA: albumin	Pre	0.037	0.070	0.062	0.054	0.008	0.64	0.18	0.04
	Post	0.092	0.129	0.109	0.181	0.019	0.09	0.01	0.39

^a Pre, prepartum (measurements made on day 7 prior to calving); Post, postpartum (overall means of day 1 and 21 postpartum); SEM, standard error of the means; HDL, high density lipoprotein; LDL, low density lipoprotein.

Table 5

Effects of supplementing with (HM) or without herbal mixture (NO) and different oil seed sources (OS; linseed or soybean) to transition dairy cows on the plasma activity of circulating enzymes (U/L) originating from the liver.

Item	Day ^a	Linseed		Soybean		SEM ^a	P values		
		HM	NO	HM	NO		OS	HM	OS × HM
AST (U/L) ^a	Pre	75.2	64.6	75.9	60.5	4.1	0.69	<0.01	0.56
	Post	75.7	75.6	74.8	75.4	3.8	0.85	0.91	0.91
ALT (U/L) ^a	Pre	93.8	96.4	104.3	90.1	6.6	0.80	0.74	0.24
	Post	77.4	73.7	81.1	77.1	7.8	0.66	0.62	0.98
ALP (U/L) ^a	Pre	120.2	105.4	130.8	114.4	16.2	0.58	0.16	0.59
	Post	126.2	121.0	126.1	130.3	15.5	0.77	0.97	0.76
GGT (U/L) ^a	Pre	24.6	27.3	19.9	25.7	3.2	0.80	0.28	0.71
	Post	29.6	33.2	24.9	46.8	5.2	0.44	0.03	0.12

^a Pre, prepartum (measurements made on day 7 prior to calving); post, postpartum (overall means of day 1 and 21 postpartum); AST, aspartate aminotransferase; ALT, alanine aminotransferase; ALP, alkaline phosphatase; GGT, γ -glutamyltransferase.

oil seed sources were observed for total antioxidant capacity during the postpartum ($P=0.09$; Table 6) and basal ($P<0.01$) and maximum conjugated dienes (CD) production ($P=0.08$; Fig. 2A) and calculated area under the curve (AUC, $P<0.01$, Fig. 2B) of CD at 1 day postpartum. The lowest basal and maximum CD were observed in cows fed HM in SB based diet. Supplementing HM resulted in greater ($P<0.01$) total antioxidant capacity during the prepartum period and lower malondialdehyde concentration than unsupplemented cows during the prepartum ($P<0.01$) and postpartum ($P=0.04$) period. Cows fed LS-diet enriched with n-3 FAs had lower total antioxidant capacity ($P=0.01$) during the prepartum period and greater malondialdehyde concentration ($P=0.02$) than cows fed SB-based diet during the postpartum period. Postpartum malondialdehyde concentration was higher in multiparous (2.03 vs. 1.20 $\mu\text{mol/L}$; $P<0.01$) than primiparous cows.

Results for IV-GTT are presented in Table 7. Basal glucose concentration ($P=0.73$), AUC ($P=0.62$), CR ($P=0.23$), time to reach half-maximal ($P=0.24$) and time to reach basal concentration of glucose ($P=0.69$) were not affected by HM supplementation. The CR of insulin to CR of glucose ratio was higher in cows fed HM than unsupplemented counterparts ($P=0.05$; Table 7). Figs. 3 and 4 illustrate basal plasma glucose and insulin and their profile change following glucose infusion, respectively. Basal glucose concentration was greater ($P=0.02$) for cows fed LS based diet than cows fed SB diet (Table 7). After glucose infusion, glucose AUC was lower ($P<0.01$) in cows fed LS-based diet. The CR of glucose ($P=0.70$), time to reach half-maximal glucose concentration ($P=0.81$) as well as the time to reach basal concentration ($P=0.37$) was not affected by different oil seed sources. There was a significant interaction ($P<0.01$)

between oil seed sources and HM supplementation on basal insulin concentration before glucose infusion as cows supplemented with HM in n-6 had greater insulin concentration than other treatments (Table 7). The CR of insulin tended to be higher in cows fed SB diet than LS diet ($P=0.10$; Table 7). Area under the curve ($P=0.99$) and time to reach half-maximal insulin concentration ($P=0.19$) were the same between two oil seed sources.

4. Discussion

Main aim of the study was to investigate the effects of supplementation with HM and different oil seed sources on metabolic profile, glucose and insulin responsiveness to glucose challenge, lipid and protein metabolism and non-enzymatic antioxidant status of transition cows.

Serum concentrations of glucose, insulin and BHBA were not affected by HM supplementation, and this metabolic effect is consistent with the findings of similar energy balance status of cows independent of HM supplementation. Similar to our finding, Tassoul and Shaver (2009) reported that supplementation of essential oil mixture to transition cows did not affect plasma concentrations of glucose and BHBA.

Results of current study showed that postpartum concentration of serum NEFA decreased in cows supplemented with HM compared to their unsupplemented counterparts, despite their similar energy balance. Although the exact mechanisms behind the lower NEFA concentration in the HM supplemented cows are unclear, our data likely suggest potential involvement of HM addition in the inhibition of lipolysis, hence enhancing lipid metabolism and improvement in metabolic status in periparturient cows

Table 6

Effects of supplementing with (HM) or without herbal mixture (NO) and different oil seed sources (OS; linseed or soybean) to transition dairy cows on serum non-enzymatic antioxidant status.

Item	Day ^a	Linseed		Soybean		SEM ^a	P values		
		HM	NO	HM	NO		OS	HM	OS × HM
TAC ^a (nmol/L)	Pre	0.52	0.44	0.58	0.53	0.02	<0.01	0.01	0.51
	Post	0.56	0.59	0.67	0.61	0.02	0.02	0.47	0.09
MDA ^a ($\mu\text{mol/L}$)	Pre	0.92	1.81	1.13	2.12	0.19	0.35	<0.01	0.78
	Post	1.45	2.37	1.18	1.38	0.25	0.02	0.04	0.17

^a Pre, prepartum (measurements made on day 7 prior to calving); Post, postpartum (overall means of day 1 and 21 postpartum); SEM, standard error of the means; TAC, total antioxidant capacity of serum; MDA, malondialdehyde.

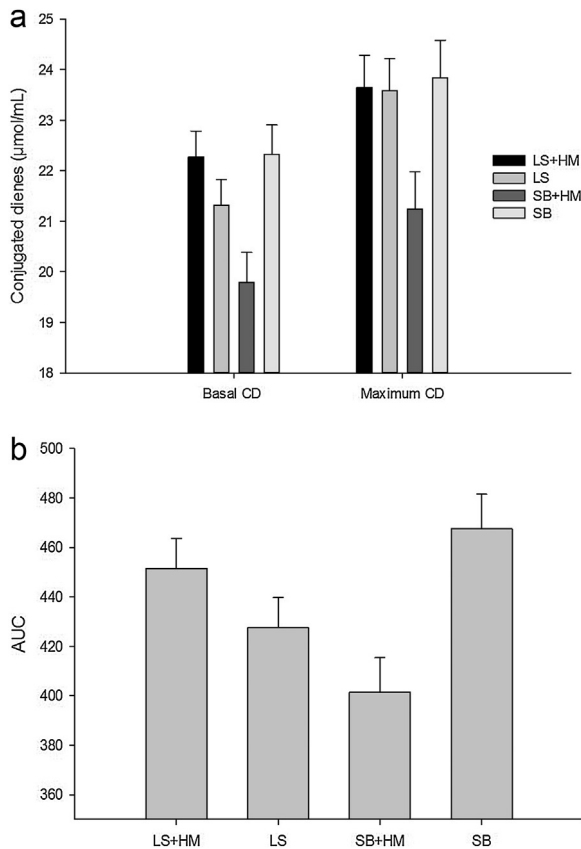


Fig. 2. (A) Effects of supplementing herbal mixture (HM) and different oil seed sources (OS; linseed, LS or soybean, SB) to transition dairy cows on basal (P -values; OS = 0.19, HM = 0.18 and OS $\times$ HM < 0.01) and maximum conjugated dienes (CD) production (P -values; OS = 0.13, HM = 0.09 and OS $\times$ HM = 0.08) at 1 days postpartum. Data were expressed as LSM $\pm$ SEM. (B) Effects of supplementing herbal mixture (HM) and different oil seed sources (OS; linseed, LS or soybean, SB) to transition dairy cows on area under the curve (AUC) of conjugated dienes (P -values; OS = 0.70, HM = 0.13 and OS $\times$ HM < 0.01) at 1 days postpartum. Data were expressed as LSM $\pm$ SEM.

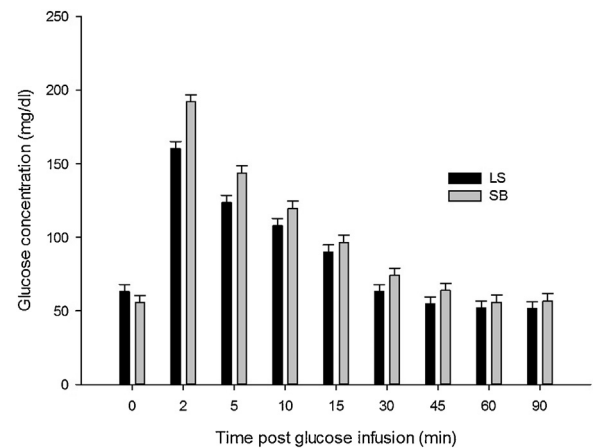


Fig. 3. Effects of supplementing different oil seed sources (OS; linseed, LS or soybean, SB) to transition dairy cows on plasma glucose concentration after i.v. glucose tolerance test (P -values for oil seed sources at times 0, 2, 5, 10, 30 and 45 min post glucose infusion were < 0.05). Data were expressed as LSM $\pm$ SEM.

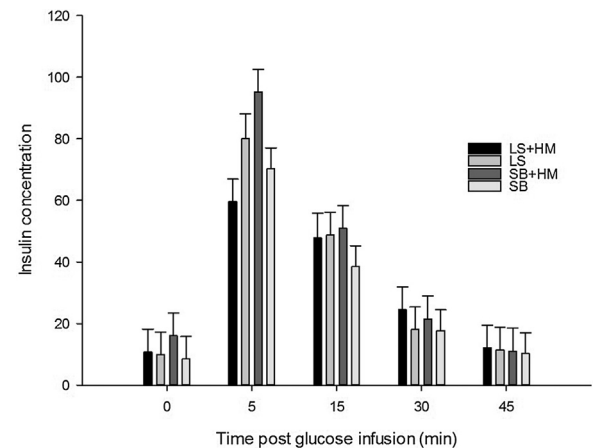


Fig. 4. Effects of supplementing herbal mixture (HM) and different oil seed sources (linseed, LS or soybean, SB) to transition dairy cows on plasma insulin concentration after i.v. glucose tolerance test. Data were expressed as LSM $\pm$ SEM.

Table 7

Effects of supplementing with (HM) or without herbal mixture (NO) and different oil seed sources (OS; linseed or soybean) to transition dairy cows on plasma glucose and insulin responses to i.v. glucose tolerance test at 25 days postpartum.

Item	Linseed		Soybean		SEM ^a	P value		
	HM	NO	HM	NO		OS	HM	OS × HM
Glucose								
Basal level (mg/dl) ^a	66.5	61.4	54.1	57.3	3.1	0.02	0.73	0.20
CR (%/min) ^a	2.34	2.32	2.40	2.61	0.30	0.70	0.23	0.56
T _{1/2} (min) ^a	31.0	25.5	30.0	27.9	3.1	0.81	0.24	0.56
AUC (mg/dl × 45 min) ^a	925	1065	1528	1530	135	<0.01	0.62	0.63
T _{basal} (min) ^a	44.5	45.9	53.2	47.3	5.4	0.37	0.69	0.51
Insulin								
Basal level (μU/mL)	10.0	9.1	15.3	9.3	0.95	<0.01	0.01	0.01
AUC (μU/mL × 45 min)	516	592	608	494	118	0.98	0.87	0.42
CR (%/min)	4.18	4.27	5.6	4.5	0.47	0.10	0.31	0.23
T _{1/2} (min)	16.8	18.7	12.6	16.9	2.2	0.19	0.18	0.59
CR _{ins} /CR _{glu}	1.711	1.540	2.378	1.559	0.229	0.16	0.05	0.18

^a SEM, standard error of the means; CR₂, clearance rate (2–45 min); $T_{1/2}$, time to half-maximal concentration; T_{basal} , time to reach basal level and AUC, area under the curve (0–45 min).

via improvement in insulin sensitivity as will be discussed later. Our results are in agreement with the results of others reporting lowered NEFA concentration in dairy ewes supplemented with rosemary extract (Chiofalo et al., 2012), and feedlot-fed bull calves fed a commercial combination of plant extract containing carvacrol, cinnamaldehyde and capsaicin (Castillo et al., 2012). Some results obtained from this study such as lower NEFA to cholesterol and NEFA to albumin ratio, as an index for predicting pre- and postpartum disease incidence (Contreras and Sordillo, 2011; Holtenius, 1989), as well as lower GGT activity in serum, suggested that cows fed HM possibly had a slight improvement in liver function than unsupplemented cows. Higher serum albumin and lower urea-N concentrations in HM supplemented cows are in agreement with earlier studies, which reported that feeding cinnamon oil to lactating goats and curcumin, main bioactive compound of turmeric, to diabetic rats resulted in higher albumin and lower urea-N concentration than control group (Tikoo et al., 2008; Kholif et al., 2012).

Although most of the parameters obtained from IV-GTT on 21 days postpartum were not affected by HM supplementation, higher CR of insulin to CR of glucose, as an indicator of insulin sensitivity, as well as improved RQUICKI value indicated that cows supplemented with HM had greater insulin sensitivity than unsupplemented cows during the postpartum period, which might be responsible for improved performance response (Hayirli et al., 2001; Holtenius and Holtenius, 2007). The presence of an interaction between HM and oil seed sources on RQUICKI suggests a mild synergism of HM and oil seed sources on insulin responsiveness during the prepartum period. Insulin sensitizing activity might be attributed to less body fat mobilization and lower serum NEFA concentration in HM group (Roche et al., 2009). In addition, many researchers reported that herbal plants, which were used in the present study such as cinnamon bark, turmeric root, clove bud and rosemary leaves, have several beneficial effects on ameliorating insulin resistance and glucose metabolism (Nishiyama et al., 2005; Sheng et al., 2008; Qin et al., 2010). Recently, it has been reported that duodenal supplementation of quercetin in cows resulted in changes in glucose metabolism, indicating increased insulin release and insulin sensitivity (Gohlke et al., 2013). These authors suggested that quercetin might assist in the adaptation of high-yielding cows to lactation.

Some possible modes of action of herbs on insulin sensitizing activity include: increasing peroxisome proliferator-activated receptor- γ activity, increasing insulin receptors and improving their function; increasing glycogen synthase activity and glycogen accumulation; attenuating inflammatory cytokine expression, accelerating fatty acid oxidation as well as increasing antioxidant status (Anderson, 2008; Shao et al., 2012). Our results for the first time showed that HM addition positively affect metabolic profile and glucose metabolism in transition dairy cows.

Cows fed LS diet had greater DM and energy intake and consequently showed more positive energy balance and lower BW loss at periparturient period. Due to better energy status, these cows had greater serum glucose and lower NEFA concentration that indicated lower body

fat mobilization during the pre- and postpartum period. Cows fed SB-based diet without HM had the lowest DMI and highest BCS change during postpartum period which resulted to greatest serum NEFA concentration at postpartum. Petit et al. (2007) reported that NEFA concentration of multiparous cows fed flaxseed had lower NEFA concentration after calving than those supplemented with Energy Booster. In addition, it has been reported that plasma NEFA concentration was lower for mid-lactating cows fed flaxseed compared with those fed sunflower seed (Petit, 2003). Lower insulin to glucose ratio as well as results obtained from IV-GTT such as a decline in glucose AUC and no change in insulin AUC in cows fed LS diet indicated that these cows had facilitated insulin action. Thus, another explanatory mechanism for observing lower serum NEFA concentration might be attributed to the more insulin responsiveness in addition to more energy intake of LS fed cows. On the other word, more insulin responsiveness of LS-fed cows may partially attributed to improved energy balance and lower body fat mobilization (Pires et al., 2008). In addition, it has been reported that abomasal infusion of n-3 FA sources such as camelina and linseed oil were effective in altering insulin responsiveness and consequently decrease the level of plasma NEFA (Pires et al., 2008; Salin et al., 2012). To the best of our knowledge, no experiments have been published examining the roles that extruded oil seeds such as linseed and soybean, as the most available sources of n-3 and n-6 FA in ruminant feeding, respectively, play during the transition period in dairy cows.

Oxidative stress (imbalance between oxidants and antioxidants; Gobert et al., 2009), leads to peroxidative damage of lipids and other macromolecules. It has been reported that during the transition period cows can experience oxidative stress, which is involved in the etiology of several diseases and disorders in this critical time (Turk et al., 2008). Our results showed that cows fed HM had higher antioxidant capacity and lower malondialdehyde as biomarker for oxidative stress than unsupplemented cows. The interactions observed between oil seed sources and HM addition in the present study indicates that supplemented HM was effective to prevent lipid peroxidation more effectively in SB-diets enriched with n-6 FAs than those fed LS-diets enriched with n-3 FAs. Greater milk production in cows fed HM in SB-based diet compared to those fed SB-diet without HM might be partially attributed to improved antioxidant status. Wang et al. (2010) reported that milk production increased in cows fed antioxidant supplement in low saturated fatty acid-based diet.

In the present study, HM provided high amounts of phenolic compounds. Previously, it has been reported that plant extracts rich in polyphenols are a good source of antioxidants to lower plasma lipid peroxidation in ruminants fed 18:3 n-3-rich diet (Gobert et al., 2009).

In spite of insulin sensitizing action of cows fed LS-based diet, these cows were more sensitive to oxidative stress evidenced by lower TAC and higher malondialdehyde concentration during the pre- and postpartum period as well as higher CD production at parturition day. Enhanced cellular metabolism due to higher milk production in cows fed

LS-based diet might be responsible for observing higher oxidative stress symptoms in these cows (Castillo et al., 2003). Also, Scislowski et al. (2005) reported that steers receiving duodenal infusion of linseed oil (rich in n-3 FA) were more susceptible to peroxidation than those receiving sunflower oil (rich in n-6 FA). But contrary to our results, they showed that feeding extruded linseed did not induce oxidative stress compared to extruded sunflower seed probably due to extensive ruminal biohydrogenation of PUFA. Differences between experiments such as physiological stage of animal, and type of feed offered might have contributed to the variation.

5. Conclusions

Our research showed that there is a benefit in terms of improved energy balance and performance response, energy metabolism and insulin resistance in transition period with feeding LS-based diet rich in n-3 FA, although this effect was associated with a mild oxidative stress than cows fed SB diet. Supplementing a mixture of herbal plants including cinnamon bark, turmeric root, clove bud and rosemary leaves increased DMI and milk production and had a modest favorable effect on body reserves mobilization and blood non-enzymatic antioxidant status more profoundly in cows fed SB-based diet. Also, supplementation of HM may improve insulin responsiveness in periparturient dairy cows.

Conflict of interest statement

None of the authors had a personal or financial conflict of interest.

Acknowledgments

The authors acknowledge the staff of Lavark Research Station for animal care, and the staff of animal nutrition laboratory for the technical assistance at the Isfahan University of Technology. We appreciate of M. Javanshir Afshar director of Supa Company (Karaj, Iran) for kindly providing catheters. The assistance of the following students in Isfahan University of Technology is also appreciated: M. Hosseini-Ghaffari, H. Akbarizadeh, H. Omid, H. Ahangaran, E. Ghasemi, S. Kargar, A. Bakhtari, A. Ebne-Ali, M.T. Karimi, and H. Fasihi. Also, we thank to Dr A. Hayirli and H. Sadri (University of Tabriz, Tabriz, Iran) for their helpful review of the manuscript.

References

Akilen, R., Tsiami, A., Devendra, D., Robinson, N., 2012. Cinnamon in glycaemic control: systematic review and meta analysis. *Clin. Nutr.* 31, 609–615.

Anderson, R.A., 2008. Chromium and polyphenols from cinnamon improve insulin sensitivity. *Proc. Nutr. Soc.* 67, 48–53.

Association of Official Analytical Chemists, 1990. *Official Methods of Analysis*, 15th ed. AOAC, Arlington, VA, USA.

Bamdad, F., Kadivar, M., Keramat, J., 2006. Evaluation of phenolic content and antioxidant activity of Iranian caraway in comparison with clove and BHT using model systems and vegetable oil. *Int. J. Food Sci. Technol.* 41, 20–27.

Bell, A.W., 1995. Regulation of organic nutrient metabolism during transition from late pregnancy to early lactation. *J. Anim. Sci.* 73, 2804–2819.

Castillo, C., Hernandez, J., Lopez-Alonso, M., Miranda, M., Benedito, J.L., 2003. Values of plasma lipid hydroperoxides and total antioxidant status in healthy dairy cows: preliminary observations. *Arch. Tierz.* 46, 227–234.

Castillo, C., Benedito, J.L., Vázquez, P., Pereira, V., Méndez, J., Sotillo, J., Hernández, J., 2012. Effects of supplementation with plant extract product containing carvacrol, cinnamaldehyde and capsaicin on serum metabolites and enzymes during the finishing phase of feedlot-fed bull calves. *Anim. Feed Sci. Tech.* 171, 246–250.

Chiofalo, V., Liotta, L., Fiumanò, R., Riolo, E.B., Chiofalo, B., 2012. Influence of dietary supplementation of *Rosmarinus officinalis* L. on performances of dairy ewes organically managed. *Small Rumin. Res.* 104, 122–128.

Contreras, G.A., Sordillo, L.M., 2011. Lipid mobilization and inflammatory responses during the transition period of dairy cows. *Comp. Immunol. Microbiol. Infect. Dis.* 34, 281–289.

Drackley, J.K., 1999. Biology of dairy cows during the transition period: the final frontier? *J. Dairy Sci.* 82, 2259–2273.

Frankic, T., Voljc, M., Salobir, J., Rezar, V., 2009. Use of herbs and spices and their extracts in animal nutrition. *Acta Agric. Slov.* 94, 95–102.

Friedewald, W.T., Levy, R.T., Fredrickson, D.S., 1972. Estimation of the concentration of low-density lipoprotein cholesterol in plasma without use of the preparative ultracentrifuge. *Clin. Chem.* 226, 499–502.

Gladine, C., Rock, E., Morand, C., Bauchart, D., Durand, D., 2007. Bioavailability and antioxidant capacity of plant extracts rich in polyphenols, given as a single acute dose, in sheep made highly susceptible to lipoperoxidation. *Br. J. Nutr.* 98, 691–701.

Gobert, M., Martin, B., Ferlay, A., Chilliard, Y., Graulet, B., Pradel, P., Bauchart, D., Durand, D., 2009. Plant polyphenols associated with vitamin E can reduce plasma lipoperoxidation in dairy cows given n-3 polyunsaturated fatty acids. *J. Dairy Sci.* 92, 6095–6104.

Gohlke, A., Ingelmann, C.J., Nürnberg, G., Weitzel, J.M., Hammon, H.M., Görs, S., Starke, A., Wolfram, S., Metges, C.C., 2013. Influence of 4-week intraduodenal supplementation of quercetin on performance, glucose metabolism, and mRNA abundance of genes related to glucose metabolism and antioxidative status in dairy cows. *J. Dairy Sci.* 96, 6986–7000.

Goli, S.A.H., Sahafi, S.M., Rashidi, B., Rahimmalek, M., 2013. Novel oilseed of *Dracocephalum kotschy* with high n-3 to n-6 polyunsaturated fatty acid ratio. *Ind. Crop. Prod.* 43, 188–193.

Hayirli, A., Bremmer, D.R., Bertics, S.J., Socha, M.T., Grummer, R.R., 2001. Effect of chromium supplementation on production and metabolic parameters in periparturient dairy cows. *J. Dairy Sci.* 84, 1218–1230.

Hayirli, A., Keisler, D.H., Doepel, L., 2011. Peripartum responses of dairy cows to prepartal feeding level and dietary fatty acid source. *J. Dairy Sci.* 94, 917–930.

Holtenius, P., 1989. Plasma lipids in normal cows around partus and in cows with metabolic disorders with and without fatty liver. *Acta Vet. Scand.* 30, 441–445.

Holtenius, P., Holtenius, K., 2007. A model to estimate insulin sensitivity in dairy cows. *Acta Vet. Scand.* 49, 29–31.

Iranian Council of Animal Care, 1995. *Guide to the Care and Use of Experimental Animals*, 1. Isfahan University of Technology, Isfahan, Iran.

Kholif, S., Morsy, T., Abdo, M., Matloup, O., El-Ella, A.A.A., 2012. Effect of supplementing lactating goats rations with garlic, cinnamon or ginger oils on milk yield, milk composition and milk fatty acids profile. *J. Life Sci.* 4, 27–34.

Kuroda, M., Mimaki, Y., Ohtomo, T., Yamada, J., Nishiyama, T., Mae, T., Kishida, H., Kawada, T., 2012. Hypoglycemic effects of clove (*Syzygium aromaticum* flower buds) on genetically diabetic KK-Ay mice and identification of the active ingredients. *J. Nat. Med.* 66, 394–399.

Lessard, M., Gagnon, N., Godson, D.L., Petit, H.V., 2004. Influence of parturition and diets enriched in n-3 or n-6 polyunsaturated fatty acids on immune response of dairy cows during the transition period. *J. Dairy Sci.* 87, 2197–2210.

Liu, R.H., 2004. Potential synergy of phytochemicals in cancer prevention: mechanism of action. *J. Nutr.* 134, 3479S–3485S.

Mahfouz, M.K.M., 2011. Curcumin improves insulin sensitivity and ameliorates serum proinflammatory cytokines levels in diabetes rat model irrespective of type of diabetes. *J. Am. Sci.* 7, 794–799.

Mashek, D.G., Grummer, R.R., 2003. Effects of long chain fatty acids on lipid and glucose metabolism in monolayer cultures of bovine hepatocytes. *J. Dairy Sci.* 86, 2390–2396.

Meagher, E.A., FitzGerald, G.A., 2000. Indices of lipid peroxidation in vivo: strengths and limitations. *Free Radic. Biol. Med.* 28, 1745–1750.

Miguel, M.G., 2010. Antioxidant and anti-inflammatory activities of essential oils: a short review. *Molecules* 15, 9252–9287.

Moure, A., Cruz, J.M., Franco, D., Dominguez, J.M., Sineiro, J., Dominguez, H., Nunez, M.J., Parajo, J.C., 2001. Natural antioxidants from residual sources. *Food Chem.* 72, 145–171.

- Nishiyama, T., Mae, T., Kishida, H., Tsukagawa, M., Mimaki, Y., Kuroda, M., Sashida, Y., Takahashi, K., Kawada, T., Nakagawa, K., Kitahara, M., 2005. Curcuminoids and sesquiterpenoids in turmeric (*Curcuma longa* L.) suppress an increase in blood glucose level in type 2 diabetic KK-Ay mice. *J. Agric. Food Chem.* 53, 959–963.
- O'Fallon, J.V., Busboom, J.R., Nelson, M.L., Gaskins, C.T., 2007. A direct method for fatty acid methyl ester synthesis: application to wet meat tissues, oils, and feedstuffs. *J. Anim. Sci.* 85, 1511–1521.
- Opsomer, G., Wensing, T., Laevens, H., Coryn, M., Kruif, A.D., 1999. Insulin resistance: the link between metabolic disorders and cystic ovarian disease in high yielding dairy cows? *Anim. Reprod. Sci.* 56, 211–222.
- Petit, H.V., 2003. Digestion, milk production, milk composition, and blood composition of dairy cows fed formaldehyde treated flaxseed or sunflower seed. *J. Dairy Sci.* 86, 2637–2646.
- Petit, H.V., Palin, M.F., Doepel, L., 2007. Hepatic lipid metabolism in transition dairy cows fed flaxseed. *J. Dairy Sci.* 90, 4780–4792.
- Pires, J.A.A., Pescara, J.B., Brickner, A.E., Rio, N.S.D., Cunha, A.P., Grummer, R.R., 2008. Effects of abomasal infusion of linseed oil on responses to glucose and insulin in Holstein cows. *J. Dairy Sci.* 91, 1378–1390.
- Qin, B., Panickar, K.S., Anderson, R.A., 2010. Cinnamon: potential role in the prevention of insulin resistance, metabolic syndrome, and type 2 diabetes. *J. Diabetes Sci. Tech.* 4, 685–693.
- Roche, J.R., Friggens, N.C., Kay, J.K., Fisher, M.W., Stafford, K.J., Berry, D.P., 2009. Invited review: body condition score and its association with dairy cow productivity, health, and welfare. *J. Dairy Sci.* 92, 5769–5801.
- Salin, S., Taponen, J., Elo, K., Simpur, I., Vanhatalo, A., Boston, R., Kokkonen, T., 2012. Effects of abomasal infusion of tallow or camelina oil on responses to glucose and insulin in dairy cows during late pregnancy. *J. Dairy Sci.* 95, 3812–3825.
- Schnitzer, E., Pinchuk, I., Bor, A., Fainaru, M., Samuni, A.M., Lichtenberg, D., 1998. Lipid oxidation in unfractionated serum and plasma. *Chem. Phys. Lipids* 92, 151–170.
- Scislowski, V., Bauchart, D., Gruffat, D., Laplaud, P.-M., Durand, D., 2005. Effect of dietary n-6 and n-3 polyunsaturated fatty acids on peroxidizability of lipoproteins in steers. *Lipids* 40, 1245–1256.
- Shao, W., Yu, Z., Chiang, Y., Yang, Y., Chai, T., Foltz, W., Lu, H., Fantus, I.G., Jin, T., 2012. Curcumin prevents high fat diet induced insulin resistance and obesity via attenuating lipogenesis in liver and inflammatory pathway in adipocytes. *PLoS ONE* 7, e28784.
- Sheng, X., Zhang, Y., Gong, Z., Huang, C., Zang, Y.Q., 2008. Improved insulin resistance and lipid metabolism by cinnamon extract through activation of peroxisome proliferator-activated receptors. *PPAR Res.* 2008, 581348.
- Tassoul, M.D., Shaver, R.D., 2009. Effect of a mixture of supplemental dietary plant essential oils on performance of periparturient and early lactation dairy cows. *J. Dairy Sci.* 92, 1734–1740.
- Tedesco, D., Tava, A., Galletti, S., Tameni, M., Varisco, G., Costa, A., Steidler, S., 2004. Effects of silymarin, a natural hepatoprotector, in periparturient dairy cows. *J. Dairy Sci.* 87, 2239–2247.
- Tikoo, K., Meena, R.L., Kabra, D.G., Gaikwad, A.B., 2008. Change in post-translational modifications of histone H3, heat-shock protein-27 and MAP kinase p38 expression by curcumin in streptozotocin-induced type I diabetic nephropathy. *Br. J. Pharmacol.* 153, 1225–1231.
- Tres, A., Bou, R., Codony, R., Guardiola, F., 2009. Dietary n-6- or n-3-rich vegetable fats and α -tocopheryl acetate: effects on fatty acid composition and stability of rabbit plasma, liver and meat. *Animal* 3, 1408–1419.
- Turk, R., Juretic, D., Geres, D., Svetina, A., Turk, N., Flegar-Mestric, Z., 2008. Influence of oxidative stress and metabolic adaptation on PON1 activity and MDA level in transition dairy cows. *Anim. Reprod. Sci.* 108, 98–106.
- Van Soest, P.J., Robertson, J.B., Lewis, B.A., 1991. Methods for dietary fiber, neutral detergent fiber, and non-starch polysaccharide in relation to animal nutrition. *J. Dairy Sci.* 74, 3583–3597.
- Wang, Y.M., Wang, J.H., Wang, C., Chen, B., Liu, J.X., Cao, H., Guo, F.C., Vázquez-Añón, M., 2010. Effect of different rumen-inert fatty acids supplemented with a dietary antioxidant on performance and antioxidative status of early-lactation cows. *J. Dairy Sci.* 93, 3738–3745.
- Zachut, M., Arieli, A., Moallem, U., 2011. Incorporation of dietary n-3 fatty acids into ovarian compartments in dairy cows and the effects on hormonal and behavioral patterns around estrus. *Reproduction* 141, 833–840.